

Technostress and Sleep Quality among University Students in Indonesia: A Cross-sectional Study

Abstract

Introduction: In this study, we examined the prevalence of technostress and sleep quality using sociodemographic characteristics and a large number of participants. We also investigated the link between technostress and sleep quality in Indonesian university students. **Methods:** This cross-sectional study included 1356 Indonesian students. The Technostress Creator Questionnaire and the Pittsburgh Sleep Quality Index were used to assess the technostress levels and sleep quality. The generalized linear model was used to determine sociodemographic factors that might predict technostress levels and sleep quality. The correlation between technostress and sleep quality was investigated using the Spearman's correlation test. **Results:** The respondents reported moderate levels of technostress (3.12 ± 0.53 out of 5.00) and poor sleep quality (7.9 ± 2.83). Higher levels of technostress were associated with lower academic levels, a lower monthly allowance, a longer duration of device ownership, a larger number of devices, and long screen durations. Science, technology, engineering, and math (STEM) students experienced lower technostress than non-STEM students. Lower monthly allowances and more screen time were linked to poor sleep quality. Higher levels of technostress were significantly associated with poorer sleep quality ($P < 0.05$). **Conclusion:** Our study found that Indonesian university students experienced moderate technostress, which was related to their academic levels, allowances, study discipline, screen times, number of owned devices, and duration of device ownership. Students reported poor sleep quality, which was linked to monthly allowance and screen time. Students with higher levels of technostress reported poorer sleep quality.

Keywords: Indonesia, Pittsburgh Sleep Quality Index, sociodemographic factors, technological stress, university students

Introduction

Technostress is the inability of humans to adapt to information and communication technology (ICT) in a healthy manner.^[1] Technostress levels vary and may be associated with sociodemographic factors. For instance, people of an older age, females, and people who have achieved a higher academic level tend to report higher levels of technostress.^[2] High levels of technostress are also commonly correlated with individuals' health and well-being, including poor sleep quality.^[3,4] Technostress, due to constant connectivity and technological demands, can cause individuals to experience strain and be overwhelmed by technology, making it more difficult to achieve restful sleep, and resulting in poor sleep quality.^[3,4] Sleep quality is defined as an individual's

self-satisfaction with all aspects of the sleep experience, such as subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, the use of sleeping medications, and daytime dysfunction.^[5] Sleep quality varies and can be associated with several sociodemographic factors. Females, people of an older age, people with lower income levels, and those who use devices extensively often report poor sleep quality.^[6,7] Technostress and poor sleep quality are currently a widespread issue in digital societies and affect a wide range of people, including university students. In this digital era, the increased ICT usage in learning activities and unstructured academic lifestyle, including working late at night and spending extensive time on technology for academic purposes, can exacerbate university students' technology related stress and impaired sleep patterns.^[3,8] This condition can result in students experiencing a decline

Eno Aulia Hapsari¹,
Eneng Nunuz
Rohmatullayaly²,
Kanthi Arum
Widayati¹

¹Department of Biology,

Faculty of Mathematics and
Science, IPB University, Bogor;

²Department of Biology, Faculty
of Mathematics and Science,
Universitas Padjadjaran,
Sumedang, Indonesia

Received: 22 May, 2024.

Revised: 11 September, 2024.

Accepted: 14 October, 2024.

Published: 13 November, 2024.

ORCID:

Eno Aulia Hapsari: <https://orcid.org/0000-0001-8847-8022>;

Eneng Nunuz Rohmatullayaly:
<https://orcid.org/0000-0002-3144-796X>;

Kanthi Arum Widayati: <https://orcid.org/0000-0002-0536-5100>

Address for correspondence:

Dr. Kanthi Arum Widayati,
Gedung Biologi, Kampus IPB
Darmaga Bogor, Bogor 16680,
Jawa Barat, Indonesia.

E-mail: widayati@apps.ipb.ac.id;
kanthiarum@gmail.com

Access this article online

Website: www.healthandbehavior.com

DOI: 10.4103/shb.shb_177_24

Quick Response Code:



How to cite this article: Hapsari EA, Rohmatullayaly EN, Widayati KA. Technostress and sleep quality among university students in Indonesia: A cross-sectional study. Asian J Soc Health Behav 2024;7:197-202.

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

in their physical, psychological, and academic performance, which can ultimately lead to a decline in their academic outcomes.^[2-4]

Previous studies among Western and Eastern undergraduate students have found technological stress to be directly linked with poor sleep quality.^[3,9] In Indonesia, previous studies in several regions, such as Tegal, Kendari, Kupang, and Surakarta, have reported high rates of gadget use and an elevated prevalence of poor sleep quality among university students. Studies in Tegal and Kendari have indicated that approximately 90% of undergraduate students have been observed to engage in excessive smartphone use and exhibit symptoms of sleep disorders, such as insomnia.^[10,11] A study in Kupang has reported that undergraduate students have been found to have high levels of smartphone use intensity and relatively poor sleep quality, with a notable correlation between these two variables.^[12] Moreover, a study in Surakarta found that stress due to extent technology use worsened sleep quality among medical undergraduate students.^[13] However, such studies were typically conducted in a single area and included only undergraduate students, so the sociodemographic factors that may be associated with technostress levels and sleep quality were not explored. As such, it is crucial to gain insight into the factors that contribute to technostress and poor sleep quality and to ascertain the direct relationship between these two variables in wider study sample. It is essential to investigate this phenomenon, as a significant proportion of Indonesian students has reported that they engage with technology to a considerable extent. In this study, we, therefore, aimed to investigate the levels of technostress and sleep quality among Indonesian university students as well as the relationship between technostress and sleep quality by considering sociodemographic factors and a large study sample. We hypothesized that technostress levels and sleep quality are associated with sociodemographic factors and that technostress is associated with Indonesian students' sleep quality.

Methods

Study design

This was a cross-sectional study.

Setting

Data collection was undertaken using Google Forms from January to December 2023. The data were distributed throughout Indonesia through a number of social media platforms, including Instagram, Twitter, WhatsApp, and LINE. The survey took approximately 15 min to complete. The online questionnaire was comprised four sections: introduction page, sociodemographic data collection, Technostress Creator Questionnaire (TSC-Q), Pittsburgh Sleep Quality Index (PSQI). The initial section was the introduction page, which provided respondents with pertinent information regarding the research title, aim,

research background, and the researcher's contact details. Before answering the questionnaire, the researcher sought informed consent from each respondent. Upon obtaining consent, the respondent was able to proceed to the subsequent section of the online questionnaire. The second section pertained to sociodemographic information. The third and fourth sections constituted the questionnaire for measuring technostress and sleep quality.

Subjects/respondents

The respondents were eligible if they were: (1) active and full-time undergraduate or graduate students from universities in Indonesia, (2) who were living in Indonesia, (3) own and use gadgets (e.g., smartphone, laptop, tablets, and personal computer), (4) with the Internet access. Students enrolled in employee programs are exempt from the stated criteria. After prospective respondents fulfill the four criteria listed above, they will be given an explanation on the Google Forms link regarding the research and procedures. If the prospective respondent states that they are willing to sign informed consent, then they can start filling out the questionnaire. A total of 1454 university students consented to participate and completed the questionnaire. After the screening and exclusion of students with missing information, the final sample for analysis consisted of 1356 Indonesian university students.

Variables

The study variables were sex, age, region of residence, academic level (undergraduate or graduate), study discipline (science, technology, engineering, and math [STEM] and non-STEM), number of devices owned, monthly allowance, duration of device ownership, and time spent using a device (screen time). Technostress and sleep quality were measured using validated, publicly available instruments.

Measurement

Adaptation of the instruments into Bahasa Indonesia

Two individuals proficient in both English and Bahasa Indonesia translated all questionnaires used to assess technostress and sleep quality into the latter language. Subsequently, the text was translated back into English by two other individuals proficient in Bahasa Indonesia–English translation to guarantee that the content was an accurate reflection of the original English version. A limited number of discrepancies in meaning were identified, and those that were present were resolved through a process of consensus. All items in the Indonesian version of the questionnaires have been validated through expert judgment. A pilot test was conducted with a sample of 100 respondents. During the pilot test, no respondents indicated any issues with the questions themselves; however, some did raise concerns regarding the presentation format. These concerns were promptly addressed and resolved, and a

reliability test was conducted using Cronbach's alpha to assess the internal consistency of the tested items. In this pilot test, the TSC-Q showed Cronbach's $\alpha > 0.80$, indicating strong internal consistency. The PSQI had a Cronbach's $\alpha > 0.50$, indicating its acceptable reliability.

The technostress creator questionnaire

The TSC-Q is used to assess individuals' technostress.^[14] It consists of 23 items, which are rated from 1 (strongly disagree) to 5 (strongly agree). The average score is used to interpret the prevalence of technostress: < 3.00 (low); $3.00-3.49$ (moderate); and $3.50-5.00$ (high).^[14] The TSC-Q's reliability has been evaluated and has shown high internal consistency with a Cronbach's $\alpha > 0.80$.

The Pittsburgh Sleep Quality Index

The Pittsburgh Sleep Quality Index was used to determine the students' subjective sleep quality in the prior months.^[5] The questionnaire contains seven components: subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, the use of sleep medications, and daytime dysfunction, each of which is scored between 0 and 3, with a total possible assessment score of between 0 and 21. A total PSQI score ≤ 5 indicates good sleep quality, whereas a score > 5 denotes poor sleep quality.^[5] The Cronbach's α of the PSQI has been assessed as > 0.50 , which confirms its acceptable reliability.

Study size

No prior sample size estimation was performed.

Statistical analysis

The descriptive statistics were analyzed using the Program R (<https://www.r-project.org>). The generalized linear model (GLM) was used to identify the sociodemographic factors that may predict technostress and sleep quality. In determining the appropriate GLM model, the variance-inflation factor (VIF) was used to check for multicollinearity between the independent variables, with only variables with a VIF score < 3 being retained for the stepwise regression.^[15] Then, stepwise regression was performed to obtain a simple model of the relevant predictors, where $P < 0.05$ was considered statistically significant. Spearman's rank correlation coefficients (ρ) were calculated to determine the relationship between technostress and sleep quality.

Ethics considerations

This research was approved by the Ethical Commission of Health Research Sulawesi Tenggara, Indonesia (ref: 02/ECHR-IPHA/I/2023). The respondents were provided with information regarding the research objectives and provided their informed consent. Respondents were given assurance that the privacy of the respondent's data would not be disseminated and would only be used for research purposes. There was also a statement that gave potential

participants the freedom to skip the link obtained if they were not willing. There was no charge for sharing or filling out the questionnaire on this dare. There was a statement of freedom for respondents to withdraw from the study at any time without any consequence.

Results

This study included 1356 Indonesian university students, who ranged in age from 17 to 67 years (mean, 22.62 ± 3.59 years). The respondents' sociodemographic characteristics are presented in Table 1.

The results of this study established that the TSC-Q scores varied from 1.48 to 5.00. The results also indicated that Indonesian university students generally had moderate levels of technostress (3.12 ± 0.53 out of 5.00). The

Table 1: Sociodemographic characteristics of the respondents

Variable	Category	Number of respondents (%)
Sex	Male	274 (20.21)
	Female	1082 (79.79)
Islands	Sumatra	206 (15.19)
	Jawa	991 (73.08)
	Kalimantan	49 (3.61)
	Sulawesi	36 (2.65)
	Bali and Nusa Tenggara	36 (2.65)
	Maluku	18 (1.33)
	Papua	20 (1.47)
Domicile	Rural	296 (21.83)
	Urban	1060 (78.17)
Education level	Undergraduate	1135 (83.70)
	Graduate	221 (28.30)
Study discipline	STEM	781 (57.60)
	Non-STEM	575 (42.40)
Monthly allowance (USD=16,182.60 IDR)	$\leq 500,000$	269 (19.84)
	500,001–1,000,000	422 (31.12)
	1,000,001–3,000,000	448 (33.04)
	3,000,001–5,000,000	131 (9.66)
	$> 5,000,000$	86 (6.34)
Having partner	Yes	407 (30.01)
	No	949 (69.99)
Number of devices owned	1	177 (13.05)
	2	724 (53.39)
	3	311 (22.94)
	> 3	144 (10.62)
Duration of device ownership (years)	0–1	3 (0.28)
	2–3	40 (2.95)
	4–5	80 (5.90)
	> 5	1233 (90.87)
Time spent using a device (screen time) (h)	0–1	414 (30.53)
	2–3	329 (24.26)
	4–5	179 (13.20)
	> 5	434 (32.00)

STEM: Science, technology, engineering, and math

university students' PSQI scores varied from 1.00 to 21.00. The results also revealed that Indonesian university students' average sleep quality was poor (7.90 ± 2.83).

The variables associated with technostress and sleep quality were obtained using the GLM ($P < 0.05$). Academic level, study discipline, monthly allowance, duration of device ownership, number of devices owned, and screen time were associated with technostress [Table 2]. Undergraduate students reported higher technostress than graduate students. The STEM students had lower levels of technostress than the non-STEM students, and the students with higher monthly incomes and a longer duration of device ownership reported less technostress. The students who owned more devices and had longer screen time were more likely to experience technostress. Monthly allowance and amount of screen time were associated with sleep quality [Table 3]. A higher monthly allowance was associated with a lower PSQI score, which indicated that students with higher monthly incomes had better sleep quality ($P = 0.025$). Longer screen time was correlated with higher PSQI scores, which showed that the more time the students spent using their devices, the worse their sleep quality ($P = < 0.001$).

Based on Spearman's correlation test, a significant positive correlation was found between higher technostress and a higher PSQI score ($\rho = 0.15$, $P < 0.001$) [Figure 1], where a higher PSQI score denoted poorer sleep quality. This indicated that the students with higher technostress experienced worse sleep quality.

Discussion

The overall technostress levels of the 1356 Indonesian students included in our study were found to be moderate.

Table 2: The sociodemographic variables associated with technostress

Variable	Estimate	SE	<i>t</i>	<i>P</i>
Academic level (undergraduate)	0.172	0.041	4.186	<0.001*
Study discipline (STEM)	-0.150	0.028	-5.273	<0.001*
Monthly allowance	-0.027	0.011	-2.354	<0.001*
Duration of device ownership	-0.120	0.033	-3.573	<0.001*
Number of devices owned	0.060	0.016	3.606	<0.001*
Screen time	0.045	0.011	3.922	<0.001*

*Generalized linear model. A $P < 0.05$ indicates statistical significance. STEM: Science, technology, engineering, and math, SE: Standard error

Table 3: The sociodemographic variables associated with sleep quality

Variable	Estimate	SE	<i>t</i>	<i>P</i>
Monthly allowance	-0.126	0.056	-2.244	0.025*
Screen time	0.232	0.062	3.720	<0.001*

*Generalized linear model. A $P < 0.05$ indicates statistical significance. SE: Standard error

The undergraduate students reported higher technostress than the graduate students. Students with lower educational attainment may experience more stress and anxiety when learning about the ICT used in lectures.^[16] The complexity of academic demands, the nature of students' studies, technological requirements, and technological invasion at the undergraduate level may contribute to this difference. The structured nature of undergraduate programs, which are frequently characterized by a higher volume of coursework and assignments, compared to graduate students, may contribute to increased technostress. Undergraduate students may also encounter digital learning environments and technology-enhanced learning for the first time. In addition, the curricula of undergraduate programs in Indonesia are frequently wide-ranging and utilize a wide range of complex technologies for a variety of subjects. As a result, students may feel uncertain about the appropriate technology to use for each course. Furthermore, undergraduate students in Indonesia may experience higher technostress levels than graduate students because of the wide variety of classes they take and their resultant interactions with high numbers of classmates, which may lead them to feel insecure about whether their fellow students have better technological skills.

The students in STEM programs had lower technostress scores than those in non-STEM programs. The different learning styles of the disciplines could explain this. We assumed that students with a background in science and technology would have stronger preferences for and greater familiarity with technology-based learning styles than those with a social background. As a result, they may be more adaptable when handling technological stressors.^[17] Students with higher monthly allowances may experience less technostress because they have easier access to better quality and convenient technology, which may their lower technology-related anxiety and enable them to cope better with technostress.^[18] Higher levels of technostress were also reported by the students who owned more devices. An increase in an individual's opportunities to experience processed information increases with the number of devices

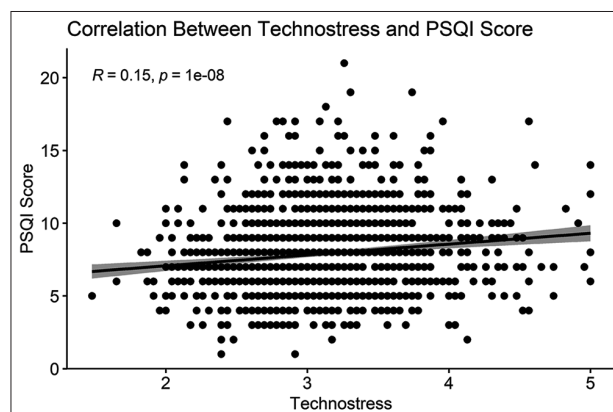


Figure 1: Correlation between technostress and sleep quality

Table 4: Variance-inflation factor report results for the generalized linear model of sociodemographic variables associated with technostress

Variable	GVIF	Df	GVIF ^{1/(1/2*Df)}
Age	1.712496	1	1.308624
Sex	1.043357	1	1.021449
Island	1.215089	6	1.016367
Domicile	1.052563	1	1.025945
Academic level	1.625969	1	1.275135
Study discipline	1.069443	1	1.034139
Monthly allowance	1.386984	1	1.177703
Duration of device ownership	1.071128	1	1.034953
Screen time	1.066911	1	1.032914
Using gadget for work	1.025758	1	1.012797
Number of device owned	1.190219	1	1.090971

GVIF: Generalized variance-inflation factor

Table 5: Variance-inflation factor report results for the generalized linear model of sociodemographic variables associated with sleep quality

Variable	GVIF	Df	GVIF ^{1/(1/2*Df)}
Age	2.010867	1	1.41805
Sex	1.040633	1	1.020114
Island	1.157946	6	1.012296
Domicile	1.041375	1	1.020478
Academic level	1.937921	1	1.392092
Study discipline	1.071212	1	1.034994
Monthly allowance	1.309806	1	1.144468
Duration of device ownership	1.060579	1	1.029844
Screen time	1.043394	1	1.021467

GVIF: Generalized variance-inflation factor

they own. Multiple devices can result in technology overload and continual exposure, which may contribute to technostress.^[19]

Similar to previous studies, the students in our study who had owned a device for a longer period had lower technostress.^[2,18] Students who have more experience using mobile devices may find that using their devices are less stressful. Such students may be more familiar and comfortable with their devices, which would lessen the stress associated with technology use. In line with a previous study, the students in our cohort who used their devices for long time experienced higher technostress.^[19] Extended screen time over the long term can lead to a persistent sense of connectedness and expose individuals to an overload of information, both of which are considered factors that can contribute to technostress.

In terms of sleep quality, the average PSQI score among the university students in the present study was 7.9 ± 2.83 , which indicated poor sleep quality. However, the students with higher monthly allowances tended to have better sleep quality, a finding that was in line with the results of a previous study.^[7] A lower monthly allowance may cause

financial strain, which may lead to difficulties in relaxing before bed and falling asleep. Higher monthly allowances may also give students the opportunity of purchasing better sleep-promoting resources, which may improve their sleep hygiene. We found that the longer students used their devices, the worse their sleep quality. Excessive screen time can disturb humans' circadian rhythms and suppress melatonin release, which can cause sleep disturbances and difficulties in falling asleep.^[9,20]

We noted a negative correlation between high technostress and poorer sleep quality. Technostress can lead to poor sleep quality due to a lack of sleep opportunities and insufficient recovery, which results in impaired biological functions, including the poor regulation of melatonin, the hormone responsible for regulating the sleep cycle.^[3,9,20] Furthermore, technostress can cause information overload, which can result in issues with falling asleep and staying asleep.^[3]

The variance-inflation factor (VIF) report results for the generalized linear model of sociodemographic variables associated with technostress and sleep quality can be seen on Tables 4 and 5.

Limitations

The cross-sectional approach of the study design limited the availability of information about the causal relationships between technostress, sleep quality, and the sociodemographic factors we investigated. Longitudinal studies may provide a better understanding of the causal links between these variables.

Conclusion

The university students in our study experienced moderate levels of technostress and poor sleep quality. Sociodemographic factors, such as education level, monthly allowances, study discipline, screen time, the number of devices owned, and the duration of device ownership, were associated with technostress levels. Monthly allowance and screen time were predictors of sleep quality. Furthermore, higher technostress was correlated with poor sleep quality among the students.

Authors' contributions

Eno Aulia Hapsari: design, definition of intellectual context, literature search, data acquisition, data analysis, statistical analysis, manuscript preparation, manuscript editing. Eneng Nunuz Rohmatullayaly: concepts, design, definition of intellectual context, manuscript preparation, manuscript editing, manuscript review. Kanthi Arum Widayati: concepts, design, definition of intellectual context, data analysis, manuscript preparation, manuscript editing, manuscript review.

Data availability statement

The article contains the study's original data; any additional inquiries can be directed to the corresponding author.

Acknowledgment

We express our gratitude to the IPB University Fast-Track Scholarship Program 2022 for their support in supporting the author's educational expenses and all respondents who have participated in this study.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. Brod C. Managing technostress: Optimizing the use of computer technology. *Pers J* 1982;61:753-7.
2. Essel H, Vlachopoulos D, Tachie-Menson A, Johnson E, Ebeheakey AK. Technology-induced stress, sociodemographic factors, and association with academic achievement and productivity in Ghanaian higher education during the COVID-19 pandemic. *Information* 2021;12:1-18.
3. Yao N, Wang Q. Technostress from smartphone use and its impact on university students' sleep quality and academic performance. *Asia Pac Edu Res* 2022;1:1-10.
4. Salo M, Pirkkalainen H, Koskelainen T. Technostress and social networking services: Explaining users' concentration, sleep, identity, and social relation problems. *Info Syst J* 2019;29:408-35.
5. Buysse DJ, Reynolds CF 3rd, Monk TH, Berman SR, Kupfer DJ. The Pittsburgh Sleep Quality Index: A new instrument for psychiatric practice and research. *Psychiatry Res* 1989;28:193-213.
6. Madrid-Valero JJ, Martínez-Selva JM, Ribeiro do Couto B, Sánchez-Romera JF, Ordoñana JR. Age and gender effects on the prevalence of poor sleep quality in the adult population. *Gac Sanit* 2017;31:18-22.
7. Güneş Z, Arslantaş H. Insomnia in nursing students and related factors: A cross-sectional study. *Int J Nurs Pract* 2017;23:10.1111/ijn.12578.
8. Pourahmad Z, Koc H. Analyzing the influence of technostress on students: A systematic literature review. *CS IT Conf Proc* 2023;13:83-97.
9. Hershner SD, Chervin RD. Causes and consequences of sleepiness among college students. *Nat Sci Sleep* 2014;6:73-84.
10. Hastuti DT, Prastiani DB. Hubungan intensitas penggunaan smartphone dengan kualitas tidur mahasiswa program studi sarjana keperawatan STIKES Bhamada Slawi (The relationship between smartphone usage intensity and sleep quality among nursing undergraduate students at STIKES Bhamada Slawi). *J Ilmu Teknol Kesehatan (E-J)* 2019;10:73-8.
11. Lestari H, Bahar H, Muhtar NT, Takin D, Dinda, Oktaviana NW, *et al.* The smartphone use and insomnia incidence in health students. *CORE* 2022;2:106-12.
12. Firmansyah MF, Rante SD, Hutasoit RM. Hubungan kecanduan penggunaan smartphone terhadap kualitas tidur mahasiswa Fakultas Kedokteran Universitas Nusa Cendana tahun 2019 (The relationship between smartphone addiction and the sleep quality of students at the Faculty of Medicine, Nusa Cendana University in 2019). *CMJ* 2020;8:535-43.
13. Budianto P, Kirana D, Hafizhan M, Putra S, Mirawati D, Prabaningtyas H. The impact of gadget use for medical education during COVID-19 pandemic on neck pain, neck disability, and sleep quality among medical students in Indonesia. *IJPHS* 2022;11:581-8.
14. Tarafdar M, Tu Q, Ragu-Nathan BS, Ragu-Nathan TS. The impact of technostress on role stress and productivity. *JMIS* 2007;24:301-28.
15. Akinwande MO, Dikko HG, Samson A. Variance inflation factor: As a condition for the inclusion of suppressor variable(s) in regression analysis. *OJS* 2015;5:754-67.
16. Qi C. A double-edged sword? Exploring the impact of students' academic usage of mobile devices on technostress and academic performance. *BIT* 2019;38:1337-54.
17. Thongsri N, Shen L, Bao Y. Investigating academic major differences in perception of computer self-efficacy and intention toward e-learning adoption in China. *IETI* 2019;57:1-13.
18. Lauterbach AS, Tober T, Kunze F, Bussemeyer MR. Can welfare states buffer technostress? Income and technostress in the context of various OECD countries. *PLoS One* 2023;18:e0295229.
19. Ayyagari R, Grover V, Purvis R. Technostress: Technological antecedents and implications. *MIS Q* 2011;35:831-58.
20. Jung CM, Khalsa SB, Scheer FA, Cajochen C, Lockley SW, Czeisler CA, *et al.* Acute effects of bright light exposure on cortisol levels. *J Biol Rhythms* 2010;25:208-16.